

EC400: Static Optimization & Fixed Points

Solutions 2025

1. (20 marks) For $(a, b) \in \mathbb{R}^2$, consider the following quadratic form

$$Q(\mathbf{x}) = ax_1^2 + bx_2^2 + (a+b)x_3^2 + 2x_1x_3.$$

- (a) (4 marks) Identify the *symmetric matrix of coefficients* A associated with this quadratic form. That is, find the matrix A satisfying $Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$. Then, find all the principal minors of A .

Solution: (2 marks) The matrix of coefficients is:

$$A = \begin{pmatrix} a & 0 & 1 \\ 0 & b & 0 \\ 1 & 0 & a+b \end{pmatrix}$$

(2 marks) First order minors are: $M_{11} = a$, $M_{22} = b$, $M_{33} = a+b$. Second order minors are: $M_{12} = ab$, $M_{13} = \det \begin{pmatrix} a & 1 \\ 1 & a+b \end{pmatrix} = a^2 + ab - 1$, $M_{23} = b(a+b) = ab + b^2$. The third order minor is: $M_{123} = \det(A) = ab(a+b) - b$.

- (b) (12 marks) Define what it means for A to be *indefinite*, *positive semidefinite* and *negative semidefinite*. Then, find all the values for a and b such that the quadratic form $Q(\mathbf{x})$ is respectively *indefinite*, *positive semidefinite* and *negative semidefinite*.

Solution: (2 marks) A quadratic form is indefinite if $Q(\mathbf{x}) > 0$ and $Q(\mathbf{y}) < 0$ for some $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$. A quadratic form is positive semidefinite if $Q(\mathbf{x}) \geq 0$ for all $\mathbf{x} \in \mathbb{R}^n$. A quadratic form is negative semidefinite if $Q(\mathbf{x}) \leq 0$ for all $\mathbf{x} \in \mathbb{R}^n$.

(4 marks) If A is positive semidefinite, all principal minors must be non-negative. Thus:

First order minors: $a \geq 0$, $b \geq 0$, $a+b \geq 0$.

Second order minors: $ab \geq 0$, $a^2 + ab - 1 \geq 0$, $ab + b^2 \geq 0$.

Third order minor: $ab(a+b) - b \geq 0$.

Given $a, b \geq 0$, the conditions $ab \geq 0$, $a+b \geq 0$, and $ab + b^2 \geq 0$ are automatically satisfied. The remaining constraints are:

- $a \geq 0$, $b \geq 0$,
- $a(a+b) - 1 \geq 0$,
- $b(a(a+b) - 1) \geq 0$.

The second constraint requires $a > 0$ and implies the third for $b \geq 0$.

Therefore the region is: $a > 0$, $b \geq 0$, $a(a+b) \geq 1$.

- (4 marks) If A is negative semidefinite, the principal minors must alternate in sign: first order minors ≤ 0 , second order minors ≥ 0 , third order minor ≤ 0 .

First order minors: $a \leq 0$, $b \leq 0$, $a+b \leq 0$.

Second order minors: $ab \geq 0$, $a^2 + ab - 1 \geq 0$, $ab + b^2 \geq 0$.

Third order minor: $ab(a+b) - b \leq 0$, which gives $b(a(a+b) - 1) \leq 0$.

Given $a, b \leq 0$, we automatically have $ab \geq 0$, $a+b \leq 0$, and $ab + b^2 \geq 0$. For the remaining constraints:

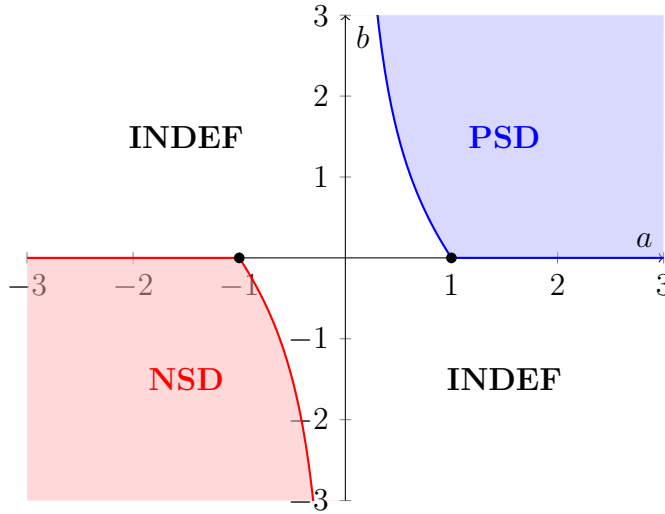
- $a \leq 0$, $b \leq 0$,
- $a(a+b) - 1 \geq 0$,
- $b(a(a+b) - 1) \leq 0$.

The second constraint requires $a < 0$ and implies the third for $b \leq 0$.

Therefore the region is: $a < 0$, $b \leq 0$, $a(a+b) \geq 1$.

(2 marks) $Q(\mathbf{x})$ is indefinite whenever it is neither positive nor negative semidefinite.

- (c) (4 marks) Plot the pairs (a, b) for which $Q(\mathbf{x})$ is respectively *indefinite*, *positive semidefinite* and *negative semidefinite* in $\mathbb{R}^2$.



2. (20 marks) Consider the function $f(\mathbf{x}) = x_1 x_2 e^{-x_1}$ defined on some convex set U in $\mathbb{R}_{++}^2$.

- (a) (6 marks) Define what it means for f to be *quasi-concave* and *quasi-convex*.

Solution: The function f is: (3 marks) *quasi-concave* on U if $C_k^+ = \{\mathbf{x} \in U | f(\mathbf{x}) \geq k\}$ is a convex set for every real number k in the image of f ; (3 marks) *quasi-convex* on U if $C_k^- = \{\mathbf{x} \in U | f(\mathbf{x}) \leq k\}$ is a convex set for every real number k in the image of f .

- (b) (14 marks) Is f *quasi-concave* or *quasi-convex*? Use counterexamples to show that a property fails.

Solution:

(7 marks) The function f is quasi-concave. To show this, consider the function $g(x_1, x_2) = \log(x_1 x_2 e^{-x_1}) = \log x_1 + \log x_2 - x_1$.

The function g is concave since: $\log x_1$ is concave for $x_1 > 0$, $\log x_2$ is concave for $x_2 > 0$, $-x_1$ is concave (linear), and the sum of concave functions is concave.

Since g is concave and $f(x_1, x_2) = e^{g(x_1, x_2)}$, and since the exponential function is a strictly increasing transformation, f is quasi-concave.

(7 marks) The function f is not quasi-convex. To show this with a counterexample, consider points $\mathbf{x} = (1, e)$ and $\mathbf{y} = (k, e^k/k)$ for $k > 1$. Observe that:

$$f(\mathbf{x}) = 1 \cdot e \cdot e^{-1} = 1 \text{ and } f(\mathbf{y}) = k \cdot e^k/k \cdot e^{-k} = 1.$$

So both points belong to the lower contour set $C_1^- = \{(x_1, x_2) : x_1 x_2 e^{-x_1} \leq 1\}$.

However, consider the midpoint $\mathbf{z} = (\mathbf{x} + \mathbf{y})/2 = ((1+k)/2, (e + e^k/k)/2)$. We have

$$\begin{aligned} f(\mathbf{z}) &= (1+k)/2 \cdot (e + e^k/k)/2 \cdot e^{-(1+k)/2} = \\ &= (1+k)/4 \cdot e^{(1-k)/2} + (1+k)/4k \cdot e^{(k-1)/2} > (1+k)/4k \cdot e^{(k-1)/2} > e^{(k-1)/2}/4 > 1, \end{aligned}$$

where the first inequality holds since $(1+k)/4 \cdot e^{(1-k)/2} > 0$, the second one since $(1+k)/k > 1$, and the third one since $e^{(k-1)/2} > 4$ for k large. Because $\mathbf{z} \notin C_1^-$ for k large, this lower contour set is not convex, violating quasi-convexity.

3. (40 marks) Consider the problem:

$$\max_{x,y} x^2y \text{ subject to } 2x + y \leq a, \ y \geq 0, \ x \geq \frac{1}{a}.$$

- (a) (10 marks) For what values of $a > 0$ is Slater condition satisfied? For such values of a , do Kuhn-Tucker necessary first order conditions identify a solution to the constrained optimization problem?

Solution: (5 marks) To check whether Slater condition holds, we need to find values of $a > 0$ such that the interior of the constraint set is non-empty. For the constraints to be compatible, we need a point (x, y) with $x > \frac{1}{a}$, $y > 0$, and $2x + y < a$. The binding case gives us:

$$2 \cdot \frac{1}{a} + 0 < a \Rightarrow \frac{2}{a} < a \Rightarrow a^2 > 2 \Rightarrow a > \sqrt{2}.$$

So Slater condition is satisfied for $a > \sqrt{2}$.

(5 marks) For $a > \sqrt{2}$, Kuhn-Tucker necessary conditions identify a solution to the constrained optimization problem, since all constraints are linear (hence convex), and the constraint qualification (Slater condition) holds.

- (b) (6 marks) Write down the Kuhn-Tucker necessary first order conditions that must be satisfied by the solution of the constrained optimization problem.

Solution: (6 marks) The Lagrangian is $\mathcal{L}(x, y|\boldsymbol{\lambda}) = x^2y - \lambda_1(2x + y - a) - \lambda_2(-y) - \lambda_3(-x + 1/a)$. The necessary first order conditions are:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial x} &= 2xy - 2\lambda_1 + \lambda_3 = 0, & \frac{\partial \mathcal{L}}{\partial y} &= x^2 - \lambda_1 + \lambda_2 = 0, \\ \lambda_1(2x + y - a) &= 0, & \lambda_2(-y) &= 0, & \lambda_3(-x + 1/a) &= 0, \\ 2x + y - a &\leq 0, & y &\geq 0, & x - 1/a &\geq 0, \\ \lambda_1, \lambda_2, \lambda_3 &\geq 0. \end{aligned}$$

- (c) (18 marks) Solve the constrained optimization problem for all values of a satisfying the restrictions found in part (a).

Solution: (6 marks) First observe that $y > 0$ at the optimum when $a > \sqrt{2}$. This follows because for $f(x, y) = x^2y$, we have that $f(x, 0) = 0$ for any $x \geq 1/a$ while $f(1/a, a - 2/a) > 0$. Since $(1/a, a - 2/a)$ fulfils all the constraints, we can conclude that $\lambda_2 = 0$ at the optimum. Moreover, $2x + y = a$ at the optimum, because the objective function is strictly increasing both variables for $y > 0$ and $x > 1/a$. Thus, $\lambda_1 > 0$.

Now, consider the two remaining cases for $a > \sqrt{2}$:

(6 marks) **Case 1:** $\lambda_1 > 0$, $\lambda_2 = 0$, $\lambda_3 = 0$. By complementary slackness, we have $2x + y = a$, $y > 0$, and $x > \frac{1}{a}$. From the FOCs, we further find:

$$\begin{aligned} 2xy - 2\lambda_1 &= 0 \Rightarrow \lambda_1 = xy, \\ x^2 - \lambda_1 &= 0 \Rightarrow \lambda_1 = x^2. \end{aligned}$$

Therefore $xy = x^2$, which gives $y = x$. Substituting into the budget constraint: $2x + x = a$, so $x = y = \frac{a}{3}$. This solution is valid when $\frac{a}{3} > \frac{1}{a}$, which requires $a > \sqrt{3}$.

(9 marks) **Case 2:** $\lambda_1 > 0$, $\lambda_2 = 0$, $\lambda_3 > 0$. By complementary slackness for the two binding constraints, we have that $x = \frac{1}{a}$, and $y = a - \frac{2}{a}$. From the FOCs, we further find:

$$\frac{2y}{a} = 2\lambda_1 - \lambda_3, \quad \frac{1}{a^2} = \lambda_1.$$

Further $\lambda_3 = 6/a^2 - 2 \geq 0$ if and only if $a \leq \sqrt{3}$, which can hold since $a > \sqrt{2}$.

Summarizing $(x^*, y^*) = (\frac{1}{a}, a - \frac{2}{a})$ for $\sqrt{2} < a \leq \sqrt{3}$, and $(x^*, y^*) = (\frac{a}{3}, \frac{a}{3})$ for $a > \sqrt{3}$.

- (d) (6 marks) Suppose the restrictions on a found in part (a) hold. Use the Envelope Theorem to express the slope of value function with respect to the parameter a in terms of the solution $(x(a), y(a), \boldsymbol{\lambda}(a))$ found in part (c).

Solution: (6 marks) The Envelope Theorem states that:

$$\frac{dv(a)}{da} = \frac{\partial \mathcal{L}(x, y | \boldsymbol{\lambda}, a)}{\partial a} \bigg|_{(x, y, \boldsymbol{\lambda}) = (x(a), y(a), \boldsymbol{\lambda}(a))} = \lambda_1(a) + \frac{\lambda_3(a)}{a^2} > 0.$$

4. (20 marks) Consider the correspondence $\mathbf{h} : [1, 4]^2 \rightrightarrows [1, 4]^2$, where $\mathbf{h}(\mathbf{x}) = (h_1(\mathbf{x}), h_2(\mathbf{x}))$ and

$$h_1(\mathbf{x}) = \begin{cases} [1, 3] & \text{if } x_2 \in [1, 2) \\ [1, 4] & \text{if } x_2 = 2 \\ [2, 4] & \text{if } x_2 \in (2, 4] \end{cases} \quad \text{and } h_2(\mathbf{x}) = 5 - x_1.$$

Argue why *Kakutani fixed point theorem* applies. Then, find all the *fixed points* of this correspondence.

Solution: (4 marks) The correspondence h_1 is upper-hemicontinuous with compact convex values, and h_2 is a continuous function (hence upper-hemicontinuous with singleton values). The domain $[1, 4]^2$ is compact and convex, and the image of $\mathbf{h}$ is contained in $[1, 4]^2$. Therefore, Kakutani's Fixed Point Theorem applies, guaranteeing at least one fixed point.

(16 marks) For a fixed point, we need $(x_1, x_2) \in \mathbf{h}(x_1, x_2) = h_1(x_1, x_2) \times h_2(x_1, x_2)$.

Since $h_2(x_1, x_2) = 5 - x_1$, at any fixed point, we must have $x_2 = 5 - x_1$.

Given the constraints $x_1 \in [1, 4]$, we have that $x_2 = 5 - x_1 \in [1, 4]$. Now we check when $5 - x_2 \in h_1(5 - x_2, x_2)$:

Case 1: If $x_2 \in [1, 2)$, then $h_1(\mathbf{x}) = [1, 3]$ and $5 - x_2 \in (3, 4]$. But if so, $5 - x_2 \notin h_1(\mathbf{x})$. Thus, there are no solutions in this case.

Case 2: If $x_2 = 5 - x_1 = 2$, then $x_1 = 3$. Further, $x_1 = 3 \in [1, 4] = h_1(3, 2)$. Thus, $(3, 2)$ is a fixed point.

Case 3: If $x_2 \in (2, 4]$, then $h_1(\mathbf{x}) = [2, 4]$ and $5 - x_2 \in [1, 3)$. But $[1, 3) \cap h_1(\mathbf{x}) \neq \emptyset$. Thus, for any $x_1 \in [1, 3) \cap [2, 4] = [2, 3)$, we have a fixed point $(x_1, 5 - x_1)$.